Web-scale research methods

Today's Topics

- Web site A/B testing
- Remote usability testing
- Online subject recruitment

Which Is Better?

NATIONAL ALERT REGISTRY

Search Results

Search Initiated | Time

Search Location | Zip | City | State

33 Registered sex offenders are located in your immediate area.

Get the complete report including map, photographs, and addresses to find out who they are and where they live. You will be able to enter your full address and specify a larger radius.

The complete report includes:

- Photos
- Appearance Details
- Conviction Information
- Offense Dates
- Offense Details
- Street Addresses
- Name Aliases

See the entire sample report

To get a complete predator report **today**, register below with National Alert Registry, and for just $19.00 we will create a custom report using your address. We will also send you a "Red Alert" notification when new predators are in your area.

Email

First Name

Last Name

Address 1

Address 2

City

State

Zip

Country

Phone

Billing Information

By submitting this information you agree to a one time charge of $19.00. Terms and Conditions

Background

The U.S. Congress passed three laws that require states to keep track of sex offenders. The Wetterling Improvements Act, the Lychner Tracking and Identification Act and Megan's Law. On March 5, 2001, The Supreme Court ruled that information about **potential predators** may be posted on the Internet.

The Problem

The information is available and you have the **right** to see it, but the process is **very difficult**. You have to know the name of the individual you are looking for and in many states you must go to your police station and request **a manual form**, or some states require a fee and **fax** to view sex offenders.

The Solution - Predator Report

Stay in the **privacy** of your home and still see the sex offenders located on a map of **your neighborhood**. The free service provides the number of predators in your neighborhood. You can then decide to purchase the full service, which includes details about these **sexual offenders**, including mobile addresses. The full service search is based on **your exact** address.

NATIONAL ALERT REGISTRY

Free Search Results for Zip Code 32225

Search Initiated: 8/2/2005 | Time: 12:12 P.M.

Search Location: State: FL | City: JACKSONVILLE, JAX | Zip: 32225

32 Registered Sex Offenders currently located in your zip code

Create Your Sex Offender Map And Detailed Predator Report

Get a custom **map of your neighborhood**, identifying your home and where offenders reside in your area. It only takes a few seconds to generate your report and costs **just $18**.

Your Neighborhood Report

Email

(Only used for RED ALERT Notifications)

First Name

Last Name

Address 1

Address 2

City

State: Select

Zip: 32225

Country: United States

Phone

Select your billing preference:

- Single "Predator Report" one time charge of $19.00
- Upgrade my registration to "Red Alert" Only $4.95/month more.

Red Alert includes:

1. **Automatically Updated Predator Reports** - Your map is updated daily when new information is available.
2. **Automatic Predator Notifications** when new offenders are in your area.
3. **Email alerts** when our nation-wide database has been updated.

Sample Report

See the entire sample report

The complete Predator Report includes:

- Color Photos (when available)
- Appearance Details
- Street Addresses
- A Detailed Map
- Names & Aliases
- Conviction Information
- Offense Dates
- Offense Details

© National Alert Registry. All rights reserved.

The designers answered this question by conducting an experiment

Half the users to their web site were randomly assigned to see one version of the page, and the other half saw the other version

users were then tracked to see how many of each actually filled the form to buy the report

revised design actually *failed* – 244 users bought the report from the original version, only 114 users bought the report from the revised version

The point is that the web site conducted a **randomized experiment** and collected data that actually tested the revision.

That's not the same as just rolling out the revised version and seeing what happens – there's a subtle but important difference. This kind of experiment is often called an **A/B test**.

A/B Testing

- A/B testing goes by other names as well
  - controlled experiment, randomized experiment, single-factor design, split test, parallel flights
- Similar approach to lab controlled experiment
  - Choose an independent variable with 2 conditions
    - e.g. the UI design to present
    - may have more than 2 conditions, e.g. A/B/C testing
  - Choose dependent variable(s) to measure
    - might be usability: time, errors, success rate
    - might be business criteria: conversions, # items bought, revenue
  - During a testing interval, randomly assign arriving users to one condition or the other
  - Do statistical testing

The distinction in web-based A/B testing is that your web site **automatically and randomly** assigns users to a condition

Ramp-up

- A/B testing can be risky
  - you're doing your testing with real users on a deployed system
  - so bugs have real consequences
- Don't go to 50/50 ratio between Control and Treatment immediately
  - Ramp up slowly: first 99.9% / 0.1%, then 99%/1%, etc.

A/A Tests

- An "experiment" that divides users into two groups with the **same** condition for both groups
  - Good for testing the experimentation infrastructure
  - You shouldn't see any difference between the groups
    - But wait! If you run 20 A/A tests and test them at the 5% significance level, then on average one of the tests will show a (phantom) significant difference
  - A/A tests also allow estimating the variance of the dependent variable
    - which is useful for power calculations

Issues with A/B Testing

- Ethics
- Predictability

when a user visits the web site, things might (randomly) be different

- Numbers, but no explanations
- Short-term vs. long-term

typical A/B test runs only for days or weeks, while the real effect of a new design might be seen only over a long term, as users learn how to use it well

But it’s worth noting that even days or weeks is a longer term than a typical lab-based user study

Recruiting Users Online

- Craigslist is a good source for lab subjects
  - for MIT subjects, the freemoney mailing list also helps
- Mechanical Turk is a labor market for tiny online tasks
  - e.g. paying $0.01 to give some keywords for an image
  - increasingly used by HCI researchers and social scientists to recruit users
- Google AdWords is another way to get users
  - generates high flow, and possibly high cost

Summary

- A/B testing offers fast, accurate testing of new web site designs in actual deployment
- Remote usability testing is getting there
- Web makes it much easier to recruit users than ever before